

# CHANG ZENG

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## EDUCATION

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|---------------------------------------------------------------------------------------|-----------------------------|
| UNIVERSITY OF MASSACHUSETTS AMHERST                                                   | Amherst, MA, 2018 - Present |
| <i>Ph.D. in Computer Science</i>                                                      | 4.000 GPA                   |
| <i>Master of Science, Major in Computer Science, Bay State Fellow</i>                 | 3.895 GPA                   |
| <i>Bachelor of Science, Major in Computer Science; Major in Environmental Science</i> | 3.832 GPA                   |

## AWARDS

- **Bay State Fellowship** at the University of Massachusetts Amherst.
- **Dean's List** recipient for the following semesters: Fall 2018, Spring 2019, Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021, and Spring 2022 at the University of Massachusetts Amherst.

## TECHNICAL SKILL

- **Programming Languages:** Python, Go, C#, C++
- **Library & Framework:** TensorFlow, PyTorch, NumPy, Pandas, Docker, OpenCV
- **Tools:** Unix/Linux, Git, Conda, Bash, AWS S3, Kubernetes, Docker, Node.js, PostgreSQL

## PUBLICATION

- CYRUS, C. CHANG, Z., AND YAIR, Z. "AI ALIGNMENT VIA POWER-MEAN ELICITATION." WORK IN PROGRESS.

## RESEARCH EXPERIENCE

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| PERSONALIZED PRODUCT SEARCH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Amherst, MA        |
| Research Assistant (Python, Recommendation System, User Features)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Dec 2024 - Present |
| <ul style="list-style-type: none"><li>• Engineered an iterative <b>self-training</b> pipeline that leverages <b>synthetic datasets</b> from multiple LLMs with <b>adaptive label smoothing</b> to boost label quality.</li><li>• Extracted and refined rich <b>user embeddings</b> and latent preferences from historical interactions using LLM-based encoding, enhanced via <b>adaptive hard-negative sampling</b> during fine-tuning.</li><li>• Developed a multi-stage fine-tuning framework to integrate user profiles into <b>personalized search</b> to jointly optimize query and item representations through <b>latent embedding</b> or <b>in-context prompt</b>.</li></ul> |                    |
| AI ALIGNMENT VIA POWER-MEAN ELICITATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Amherst, MA        |
| Research Assistant (Optimization, Algorithm, Fairness Division)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sep 2022 - Present |
| <ul style="list-style-type: none"><li>• Modeled human and LLM decision-making processes by analyzing the behavior of the <b>Weighted Generalized Means</b> class in real-world scenarios involving multiple stakeholders.</li><li>• Created flexible, robust distance <b>metrics</b> to quantify <b>welfare disparities</b> by comparing utility and disutility outcomes among diverse stakeholder groups.</li><li>• Conducted <b>minimax complexity analysis</b> to evaluate the <b>algorithm's efficiency</b> for <b>proper and improper epsilon-elicitation</b> of fairness concepts.</li></ul>                                                                                    |                    |

## WORK EXPERIENCE

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### X-CAMP TECH TEAM INTERN

Remote, US

#### Software Engineer (Go, Git, Test & Debug, CI/CD, Kubernetes, Docker)

Jun 2023 - Aug 2024

- Designed and developed a **scalable architecture** using **Golang** to seamlessly integrate Zoom API functionalities into existing systems, resulting in an improved teaching and learning experience.
- Utilized an automated **CI/CD** workflow for building and deploying to a remote **Kubernetes** cluster, facilitating seamless service scaling and management in a **containerized** environment.

## PROJECT EXPERIENCE

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### WUHUU INFORMATION SHARING PLATFORM

Amherst, MA

#### Software Engineer (Python, Crawler, AWS S3)

Sep 2022 - Oct 2023

- Developed customized **crawlers** utilizing Python to systematically gather data from social platforms, focusing on user engagement trends, content popularity, and sentiment analysis.
- Implemented a robust storage infrastructure using **AWS S3** to securely store and retrieve shared files, while optimizing data retrieval and minimizing latency.

### HEALTH DETECTION VIA HEARTBEAT ANALYSIS

Amherst, MA

#### Software Engineer (Python, Deep Learning)

Sep 2022 - Dec 2022

- Developed an autoencoder to effectively **denoise** individual heartbeats from electrocardiogram signals, enhancing signal quality and accuracy.
- Performed a comprehensive comparison between linear and non-linear structures, such as **CNN** and **RNN**, to evaluate their performance in optimizing the denoising process.

### USER STATUS CLASSIFICATION USING FACIAL TRACKING

Amherst, MA

#### Software Engineer (Python, Machine Learning, OpenCV)

Sep 2021 - Dec 2021

- Employed machine learning models, such as **BERT transformer (TensorFlow)**, with techniques such as **random forests** and **stratified k-fold**, to accurately classify facial behavior patterns.
- Utilized **OpenCV** libraries to capture and process live camera feed in real-time, extracting relevant features from the eye images, such as pupil dilation and eye movement, to determine user status.